

Erfan Nourbakhsh

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EDUCATION

Ph.D. in Computer Science, University of Texas at San Antonio (UTSA) *Jan 2025 - Present*
CGPA: 4.0 – Advisors: Dr. Anthony Rios & Dr. Ke Yang

B.Sc. in Computer Engineering, University of Isfahan *Sep 2019 - Aug 2023*
CGPA: 3.7 – Advisor: Dr. HB Kashani

RESEARCH INTERESTS

AI in Healthcare; Biomedical Informatics; Computational Social Science; Trustworthy AI; Natural Language Processing; Multi-Agent Systems; Mechanistic Interpretability; Vision Language Models (VLMs); Large Language Models (LLMs); Retrieval-Augmented Generation (RAG); Human-AI Interaction; Graph Neural Networks (GNNs); Multimodal AI

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Texas at San Antonio (UTSA) *May 2025 - Present*

- Studying the **accuracy–reasoning budget trade-off** in medical question answering, developing methods that help large language models reason with fewer tokens while achieving more accurate predictions.
- Built **MedProb**, a lightweight linear-probing framework for medical question answering that outperformed prompting-based VLMs and agentic pipelines across PATH-VQA, SLAKE, and VQA-RAD.
- Advanced wearable **cognitive load detection** by adapting 12-lead clinical ECG foundation models to 3-lead wearable ECG signals, improving macro-F1 by **11.2** and **16.1 percentage points** over from-scratch baselines.

University Instructor, University of Texas at San Antonio (UTSA) *Aug 2025 - May 2026*

- Taught *CS 2113: Fundamentals of OOP*, focusing on encapsulation, abstraction, modularity, and **polymorphism**.
- Guided students in problem analysis and coding solutions with **Java** classes and objects.
- Evaluated student performance through programming projects, quizzes, and written assessments.

Graduate Research Assistant, University of Isfahan *Sep 2023 - Dec 2024*

- Designed a low-code platform to automatically generate **chatbots**, reducing manual **development effort**.
- Automated chatbot creation from user-provided designs, **improving accessibility and usability**.
- Implemented chatbots for diverse categories (**forms, Q&A, interactive, Generative AI-Based using LLMs**).

Artificial Intelligence Specialist, Directam *Jun 2024 - Oct 2024*

- Integrated **AI models** into applications, improving scalability and reducing deployment issues.
- Developed NLP-driven systems to analyze Telegram interactions, **enhancing customer feedback analysis**.
- Built AI solutions: **Car Automatic Speech Recognition (ASR)**, Instagram Comment Generator, and **OCR-based Fortune Wheel Recognition**.

Artificial Intelligence Strategy Consultant, Pezeshke Khoob *Aug 2024 - Sep 2024*

- Led AI-focused data preparation and analysis, **reducing processing time by 50%** and accelerating insights.
- Developed intelligent **recommender systems** that improved **patient–doctor matching**, enabling patients to discover the most relevant healthcare providers based on their needs and preferences.
- Aligned **predictive modeling** with **strategic decisions**, driving data-informed insights and improving efficiency.

Machine Learning Engineer, Eftekhhar Modiran *Mar 2023 - Sep 2023*

- Developed a production-grade **ML–powered system** for job search and intelligent **candidate matching**, using classic ML algorithms such as **Random Forest** and deep learning models including **MLP** and **CNN**, leveraging **PyTorch** as the core framework.
- Collaborated with psychology and management teams to integrate **personality tests** into matching.

Undergraduate Research Assistant, University of Isfahan *Nov 2022 - Feb 2024*

- Built a document summarization system leveraging **heterogeneous graph neural networks (HGNNs)** with **BiLSTM** and **CNN** models, enhanced by **Graph Attention Networks (GATs)** for semantic relationship modeling.

- Designed a **summarization classifier** to select the most informative sentences.

Machine Learning Engineer Intern, Vitraco

Jul 2021 - Sep 2021

- Built a **recommendation system** with **PyTorch** and scikit-learn to match loads with drivers.
- Applied advanced **NLP (Transformer embeddings, topic modeling, sentiment analysis)** and **data mining** on real-world driver feedback to engineer ranking features, improving assignment **accuracy**.

Frontend Developer Intern, Safir

Aug 2020 - Jan 2021

- Architected and delivered a **ReactJS** and **TypeScript**-based platform, employing **Client-Side Rendering/Server-Side Rendering** architecture to streamline content evaluation while ensuring engineering quality standards.

PUBLICATIONS

PEER-REVIEWED CONFERENCE AND WORKSHOP PUBLICATIONS

1. **E Nourbakhsh**, K Yang, and A Rios. MedProb: Probing Internal Representations of Vision-Language Models for Medical Question Answering. In *Findings of the Association for Computational Linguistics: EMNLP 2026*.
2. **E Nourbakhsh**, MS Sirjani, A Mousavi, K Nguyen, J Quarles, M Xie, and R Slavin. 2026. Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods. In *Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (ACL 2026)*.
3. **E Nourbakhsh**, R Slavin, K Yang, and A Rios. 2026. When Retrieval Doesn't Help: A Large-Scale Study of Biomedical RAG. In *Proceedings of the 25th Workshop on Biomedical Language Processing (ACL 2026)*.
4. **E Nourbakhsh**, R Slavin, K Yang, and A Rios. 2026. TRACER: Early Failure Detection for Task-Oriented Dialogue. In *Proceedings of the 27th Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)*.
5. MS Sirjani, M Ahmad, A Mousavi, **E Nourbakhsh**, and K Nguyen. 2026. Optimizing Task Scheduling in Fog Computing with Deadline Awareness. In *Proceedings of the 2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC)*.
6. **E Nourbakhsh**, N Sanjari, and A Nourbakhsh. 2025. KD-OCT: Efficient Knowledge Distillation for Clinical-Grade Retinal OCT Classification. In *Proceedings of the International Conference on Signal Processing and Intelligent Systems (ICSPIS)*.
7. A Mazaherian and **E Nourbakhsh**. 2026. Unveiling User Perceptions in the Generative AI Era: A Sentiment-Driven Evaluation of AI Educational Apps' Role in Digital Transformation of e-Teaching. In *Proceedings of the 19th National and 13th International Conference on e-Learning and e-Teaching (ICeLeT 2026)*.
8. **E Nourbakhsh** and HB Kashani. 2024. ConHGNN-SUM: A Contextualized Heterogeneous Graph Neural Network for Extractive Text Summarization. In *Proceedings of the 20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISP)*.

PUBLICATIONS UNDER REVIEW AND IN PREPARATION

1. D Schumacher, **E Nourbakhsh**, R Slavin, and A Rios. Prompting Underestimates LLM Capability for Time Series Classification. *Under Review in Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.
2. **E Nourbakhsh**, MS Sirjani, M Xie, A Rios, and K Yang. GlobalVQA: Benchmarking Global Visual and Textual Perception in Vision-Language Models. *Under Review in Proceedings of the 2026 Advances in Neural Information Processing Systems (NeurIPS 2026 Evaluations and Datasets Track)*.
3. A Mousavi, MS Sirjani, **E Nourbakhsh**, M Xie, R Slavin, L Neely, J Davis, and J Quarles. MambaGaze: Bidirectional Mamba with Explicit Missing Data Modeling for Cognitive Load Assessment from Eye-Gaze Tracking Data. *Under Review in 2026 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*.
4. A Mousavi, **E Nourbakhsh**, MS Sirjani, M Xie, R Slavin, L Neely, J Davis, and J Quarles. CogAdapt: Transferring Clinical ECG Foundation Models to Wearable Cognitive Load Assessment via Lead Adaptation. *Under Review in 2026 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*.
5. **E Nourbakhsh**, P Mohammadi, and M Sharbaf. BINGO: A Low-Code Development Platform for Automatic Generation of Customized Chatbots for Websites. *Under Review in Journal of Systems and Software (Elsevier)*.
6. **E Nourbakhsh**, K Yang, and A Rios. ECG-DualVLM: Dual-View Vision-Language Model for ECG Clinical Understanding. *In Preparation to submit to The 41st Annual AAAI Conference on Artificial Intelligence (AAAI 2027)*.

7. MS Sirjani, **E Nourbakhsh**, A Mousavi, J Quarles, R Slavin, and M Xie. FunKAN-Lite: Hardware-Aware Functional Kolmogorov-Arnold Networks for Efficient Medical Image Enhancement and Segmentation. *In Preparation to submit to Proceedings of the 32nd Asia and South Pacific Design Automation Conference*.
8. SH Hosseini, **E Nourbakhsh**, and M Sharbaf. Vision-Language Models in Financial Forecasting: A Systematic Mapping Study. *In Preparation to submit to Journal of Frontiers in Artificial Intelligence*.

SCHOLARSHIPS & AWARDS

Awarded the Associate Certification

Center for the Integration of Research, Teaching, and Learning (CIRTL) at UTSA Jun 2026

Travel Award to Attend ACL 2026 (Award Amount: \$1,500)

Department of Computer Science, University of Texas at San Antonio (UTSA) Apr 2026

Travel Award to Attend ACL 2026 (Award Amount: \$750)

Graduate School, University of Texas at San Antonio (UTSA) Apr 2026

People's Choice Award for Outstanding and Innovative Research (Award Amount: \$250)

Graduate Student Appreciation Week, University of Texas at San Antonio (UTSA) Apr 2026

Favorite Venture Pitch Award (Award Amount: \$10,000)

Draper Data Science Business Plan Competition, University of Texas at San Antonio (UTSA) Apr 2026

Awarded the Roadrunner Teaching Assistant Academy Certificate

Division of Academic Innovation, University of Texas at San Antonio (UTSA) Feb 2026

Selected Participant, DAIR3 Bootcamp

University of Texas at San Antonio (UTSA), NIH-funded Feb 2026

Scholarship Recipient, NSF AI Spring School (MATRIX AI)

University of Texas at San Antonio (UTSA), NSF-funded Feb 2026

Awarded Certificate in Promoting Active Learning

Association of College and University Educators (ACUE) Dec 2025

Accepted Participant, ComSciCon Houston 2025

Rice University, Houston, Texas Apr 2025

PROFESSIONAL SERVICES

- **Ethics Reviewer** for The Fortieth Annual Conference on Neural Information Processing Systems, NeurIPS 2026.
- **Reviewer** for The Fifth Generation, Evaluation & Metrics Workshop, ACL 2026.
- **Reviewer** for the Ninth AAAI/ACM Conference on AI, Ethics, and Society, AAAI 2026.
- **Reviewer** for the Efficient Reasoning Workshop, COLM 2026.
- **Member, Informatics Committee** for the 2024 International Conference on e-Learning and e-Teaching.

TALKS & PRESENTATIONS

1. Oral presentation of "Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods" at the GEM Workshop session, ACL 2026, held in San Diego, California.
2. Poster presented, "Detecting Incompleteness in Feature Requests: Leveraging Codebase-Augmented LLMs in Open-Source Software," at the Texas NLP Symposium 2026, held at Texas A&M University, College Station, Texas.
3. Poster presented, "Prompting Underestimates LLM Capability for Time Series Classification," at the Texas NLP Symposium 2026, held at Texas A&M University, College Station, Texas.
4. Poster presented, "Detecting Incompleteness in Feature Requests: Leveraging Codebase-Augmented LLMs in Open-Source Software," at the NSF AI Spring School 2026, held at UT San Antonio, San Antonio, Texas.
5. Poster presented, "Detecting Incompleteness in Feature Requests: Leveraging Codebase-Augmented LLMs in Open-Source Software," at Graduate Student Appreciation Week, held at UT San Antonio, San Antonio, Texas.

TEACHING EXPERIENCE

- Graduate Teaching Assistant, CS 3443: Application Programming** May 2026 – Present
Supervised by Juan Valadez
Advanced application development course covering the software life cycle, best programming practices, and modern development tools. Supported students through the application development process, including office hours, grading, and course activities.
- University Instructor, CS 2113: Fundamentals of Object-Oriented Programming** Jan 2026 – May 2026
Designed and updated course materials, assignments, and examinations aligned with object-oriented programming learning objectives. Evaluated student performance through programming projects, quizzes, and written assessments. Provided detailed feedback to improve code quality, design practices, and logical reasoning.
- University Instructor, CS 2113: Fundamentals of Object-Oriented Programming** Aug 2025 – Dec 2025
Taught CS 2113: Fundamentals of Object-Oriented Programming, focusing on encapsulation, abstraction, modularity, and polymorphism. Guided students in problem analysis and coding solutions using Java classes and objects. Introduced computational complexity concepts to evaluate and compare algorithm efficiency.
- Graduate Teaching Assistant, CS 3733: Operating Systems** May 2025 – Aug 2025
Supervised by Dr. Chowdhury
Assisted in delivering the Operating Systems course. Held office hours, graded assignments, and supported students in understanding operating systems concepts, including process management, memory management, file systems, and concurrency.
- Graduate Teaching Assistant, CS 3733: Operating Systems** Jan 2025 – May 2025
Supervised by Dr. Korkmaz
Assisted in delivering the Operating Systems course. Conducted recitation sessions, graded homework and exams, and provided guidance to students on systems programming concepts.
- University Instructor, Computer Skills Workshop** Oct 2023 – Feb 2024
Taught the Computer Skills Workshop course for undergraduate computer engineering students, covering practical computing skills, productivity tools, and software fundamentals.
- Teaching Assistant, Basics of Computer and Programming** Oct 2023 – Jan 2024
Supervised by Dr. Shafei
Assisted in teaching introductory programming concepts to first-year students. Conducted hands-on lab sessions, graded coding exercises, and held office hours for individualized support.
- Head Teaching Assistant, Internet Engineering** Jan 2023 – Jun 2023
Supervised by Dr. Tork Ladani
Led the teaching assistant team for the Internet Engineering course. Designed and graded assignments, conducted lab sessions, and mentored students in web protocols, network architecture, and internet application development.
- Head Teaching Assistant, Data Structures** Jul 2022 – Jan 2023
Supervised by Dr. Zojaji
Led the teaching assistant team for the Data Structures course. Organized weekly problem-solving sessions, prepared supplementary materials, and evaluated student projects covering arrays, linked lists, trees, graphs, and sorting algorithms.
- Head Teaching Assistant, Operating Systems** Jul 2022 – Jan 2023
Supervised by Dr. Kaedi
Led the teaching assistant team for the Operating Systems course. Facilitated lab sessions on process management, memory allocation, and file system implementation. Graded assignments and provided academic support.
- Teaching Assistant, Systems Analysis & Design** Jan 2022 – May 2022
Supervised by Dr. Rajaei
Supported the Systems Analysis & Design course for undergraduate students. Assisted with UML modeling, requirements engineering exercises, and project evaluations.

SKILLS

Programming Languages: Python, JavaScript, Java, C, C++, Bash, TypeScript
Data Analysis & Machine Learning: Pytorch, Pytorch Geometric, Tensorflow, OpenCV, Data Analysis, Data Visualization, A/B Testing, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Apache Spark, PySpark, Airflow, Tableau, LangChain, LlamaIndex, Hugging Face Transformers, FastAPI / Flask, FAISS, Vector Databases, Diffusion Models, Data Mining, Web Scraping, LLM Optimization, PEFT (LoRA, QLoRA, ...), Synthetic Data Generation, GRPO, DPO, SFT
Software, Tools & Platforms: Object-Oriented Programming, REST APIs, Web Development, Git, GitHub, CI/CD Pipelines, Docker, Kubernetes, Linux/Unix, Agile/Scrum, AWS, Azure, MySQL, PostgreSQL, MongoDB, Jupyter, VS Code, Postman, Elasticsearch, Redis, Apache Kafka, Anaconda, LaTeX